

# Exact Concept-Grouped SHAP Explanations of a Deployed Real-Time PM2.5 Ensemble for Texas Census Tracts

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**Abstract**—Low-cost sensor networks now support operational, neighborhood-scale air quality tools, but the machine learning models behind these tools are rarely explained to the communities that rely on them. We present the explanation layer of Shared Skies, a deployed web platform that maps fine particulate matter (PM2.5) in real time across all 6,896 Texas census tracts using a tree ensemble trained on 285,798 daily readings from 310 PurpleAir sensors and validated at a leave-one-sensor-out coefficient of determination of 0.71. Because SHAP attributions are additive and the deployed prediction is a convex blend of tree models, blending per-model TreeExplainer attributions with the deployment weights reproduces the ensemble output to within one millionth of the measurement unit. Summing attributions within seven concept groups yields a decomposition non-specialists can read in physical units. The explanations show that the model is chiefly a spatial nowcaster: features describing nearby sensors dominate globally, carry a major smoke event almost alone while the dedicated smoke tier adds at most 0.31 micrograms per cubic meter, and actively suppress predictions during hyper-local events that no neighboring sensor observes. We report guardrails for policy interpretation, including inverted proximity features and strictly correlational environmental justice attributions, and show how explained failures motivate targeted sensor placement.

**Index Terms**—explainable artificial intelligence, SHAP, air quality, PM2.5, low-cost sensors, ensemble learning, environmental monitoring

## I. INTRODUCTION

Low-cost optical sensors have changed who gets to see air pollution. Networks such as PurpleAir now report fine particulate matter (PM2.5) at neighborhood scale and minute cadence, and corrected versions of their data now feed public tools such as the AirNow Fire and Smoke Map [1]. The placement of these sensors is itself uneven across income groups [2], so the models that interpolate between them are, for many communities, the only air quality information available.

Such tools are judged almost entirely on accuracy. A fused ensemble over sensor, satellite, meteorological, and demographic inputs can validate well yet behave in ways its operators would not expect, and when its outputs color a public map or steer monitor placement, unexamined behavior is a liability. Shapley additive explanation (SHAP) methods [3], [4] have made attribution practical for tree models, and environmental applications such as flood-susceptibility mapping have begun

to adopt them [5]. Yet the explanations almost always describe research models; deployed systems, the ones people actually consult, remain largely unexplained.

This paper explains one such system exactly. Shared Skies is a live web platform that predicts PM2.5 for every census tract in Texas from PurpleAir sensors and public data streams, refreshing continuously. We contribute: (1) an exact explanation layer for the deployed convex tree ensemble, which blends per-model TreeExplainer attributions with the deployment weights to reconstruct every pre-clipping output to within  $10^{-6} \mu\text{g}/\text{m}^3$ , then partitions the 38 features into seven concept groups so that each prediction becomes a short physical story; (2) a characterization of what the deployed model actually learned, which is dominated to a degree accuracy metrics never reveal by same-day readings from neighboring sensors; and (3) an explanation-driven failure analysis showing that the same neighbor features that produce the model's accuracy actively suppress its predictions during hyper-local pollution events, a structural blind spot that argues for targeted sensor placement rather than model tuning.

## II. THE DEPLOYED SYSTEM

### A. Platform

Shared Skies serves a public, bilingual web map of model-estimated 24-hour PM2.5 for all 6,896 Texas census tracts. The backend pulls live PurpleAir readings, cached for 15 minutes, recomputes the full feature set, runs the ensemble for every tract, and caches predictions for 30 minutes. A served value estimates a tract's daily-mean-equivalent PM2.5 given current conditions; the training feature pipeline is reproduced at serving time, so sub-daily dynamics enter only through the inputs. The tract color scale breaks at  $9 \mu\text{g}/\text{m}^3$ , adopting the level of the 2024 annual PM2.5 standard [6] as a fixed reference for the 24-hour map. A second view proposes future sensor placements, which Section V connects back to the explanations.

### B. Training data and features

The training target is the daily mean of the raw PurpleAir ATM-channel PM2.5 concentration. Readings are restricted to

0 to 200  $\mu\text{g}/\text{m}^3$  and cleaned per sensor with a robust z-score rule on median absolute deviation; sensors with fewer than 60 valid days, near-constant output, or medians indicating indoor placement are dropped. The result is 285,798 daily readings from 310 outdoor sensors inside Texas between January 2021 and May 2026. Sensors in border states contribute same-day neighbor context but are never training targets. The raw ATM channel is kept, rather than corrected [1], so that training targets and live inference share semantics; Section V revisits this choice. Days with means above 75  $\mu\text{g}/\text{m}^3$  are excluded from training, a choice whose consequences the explanations make visible.

Each row carries 38 features (Table I): daily weather from Open-Meteo [7]; same-day neighbor statistics (mean and count of other sensors within 25, 50, and 100 km, plus dispersion at 50 km, computed with a haversine BallTree); an ordinal smoke tier from NOAA Hazard Mapping System (HMS) analyst polygons [8]; aerosol optical depth and surface PM2.5 from the Copernicus Atmosphere Monitoring Service (CAMS) [9]; eight tract-level environmental justice indicators from EPA EJScreen [10]; location and distance covariates; and calendar encodings.

### C. Ensemble and validation

Four regressors are trained: a random forest [11], LightGBM [12], XGBoost [13], and CatBoost [14]. Validation is leave-one-sensor-out (LOSO): each of the 310 sensors is held out in turn, models are refit on the rest, and, critically, the neighbor features of every training row are recomputed with the held-out sensor removed from the pool, closing a same-day leakage path that random splits cannot detect. Spatially naive cross-validation is known to overstate the skill of models like this one [15]; under random 80/20 splitting this ensemble scores  $R^2 = 0.84$ , which LOSO deflates to 0.71.

Blend weights are a simplex-constrained convex combination fit on the LOSO out-of-fold predictions and selected by a grouped cross-fit over sensors. The deployed weights are 0.6827 (random forest), 0.3094 (CatBoost), 0.0079 (LightGBM), and exactly zero for XGBoost, giving LOSO  $R^2 = 0.7136$  against 0.7114 for an equal-weight blend and an inverse-MSE baseline at 0.7126 (RMSE 4.22  $\mu\text{g}/\text{m}^3$ , MAE 2.62  $\mu\text{g}/\text{m}^3$ ). The random forest alone reaches 0.7138: at this resolution the top candidates are tied, and the deployment is a nearly single-model instance of the convex blends the method covers. We explain the blend because the blend is what ships. Live inference fills unavailable features with training medians.

## III. EXACT ENSEMBLE EXPLANATIONS

### A. Blending per-model attributions

The deployed prediction for features  $x$  is  $f(x) = \max(0, \sum_m w_m f_m(x))$  over the three models with  $w_m > 0$ . TreeExplainer [4] decomposes each tree model exactly as  $f_m(x) = b_m + \sum_j \phi_j^m(x)$ . Attributions are additive, so

$$\sum_m w_m f_m(x) = b + \sum_j \phi_j(x) \quad (1)$$

TABLE I  
THE SEVEN CONCEPT GROUPS PARTITIONING THE 38 MODEL FEATURES  
(GROUP SIZES IN PARENTHESES).

| Group                        | Features                                                                                   |
|------------------------------|--------------------------------------------------------------------------------------------|
| Regional PM signal (9)       | neighbor PM2.5 mean and count at 25, 50, 100 km; dispersion at 50 km; CAMS AOD; CAMS PM2.5 |
| Wildfire smoke (1)           | HMS smoke tier (none/light/medium/heavy)                                                   |
| Meteorology (7)              | temperature, humidity, pressure, wind speed, precipitation, two interaction terms          |
| Local sources (4)            | EJScreen traffic, diesel PM, Superfund, and Risk Management Plan facility proximity        |
| Community and EJ context (4) | EJScreen demographic index; percent people of color, low income, linguistically isolated   |
| Geography (4)                | latitude, longitude, distance to coast, distance to nearest sensor                         |
| Season and calendar (9)      | month, weekday, day of year, and their sine/cosine encodings                               |

with  $b = \sum_m w_m b_m$  and  $\phi_j(x) = \sum_m w_m \phi_j^m(x)$  is an additive decomposition of the pre-clipping ensemble output, computed without sampling or surrogate models. Exact here means local accuracy, faithful reconstruction of this model's output, not axiomatic Shapley values under an interventional distribution: TreeExplainer (SHAP 0.50.0) runs in tree-path-dependent mode, which conditions on the training distribution through tree cover counts, and its credit assignment among correlated features differs from interventional variants. The audit of Eq. (1) runs wherever rows are explained; over the 6,000-row global sample the maximum absolute reconstruction error is  $8.9 \times 10^{-7}$   $\mu\text{g}/\text{m}^3$  (mean  $5.9 \times 10^{-8}$ ); the residual reflects float32 caching, and direct float64 explanation of individual rows reconstructs to  $5 \times 10^{-13}$   $\mu\text{g}/\text{m}^3$ . The serving layer applies only a final clip at zero, which never binds on the sampled predictions (minimum raw output +0.4  $\mu\text{g}/\text{m}^3$ ).

### B. Concept groups

Individual attributions for 38 features are exact but not legible to a public audience. We therefore partition the features into the seven concept groups of Table I and report group sums  $\Phi_G(x) = \sum_{j \in G} \phi_j(x)$ . Because the partition is exhaustive and disjoint,  $b + \sum_G \Phi_G(x)$  still reconstructs the prediction exactly; runtime validation asserts the partition against the deployed feature list. Note that a group's global importance, the mean absolute summed contribution, falls below the sum of member importances when members cancel.

### C. Analysis protocol

Global structure is computed over a uniform 6,000-row sample of the training frame (seed 42). Feature response is analyzed with dependence scatters, 12-quantile binned medians, and Spearman rank correlations between feature value and attribution, with bootstrap intervals clustered by sensor. Because the sample contains only five heavy-smoke rows, tier-level smoke statistics are recomputed exactly over all 431 heavy-tier rows in the corpus and a 2,500-row sample of the medium tier, with 2,000-resample percentile-bootstrap 95% confidence intervals. Two full days are explained sensor-by-sensor: the strongest smoke day (May 27, 2024) and the cleanest day

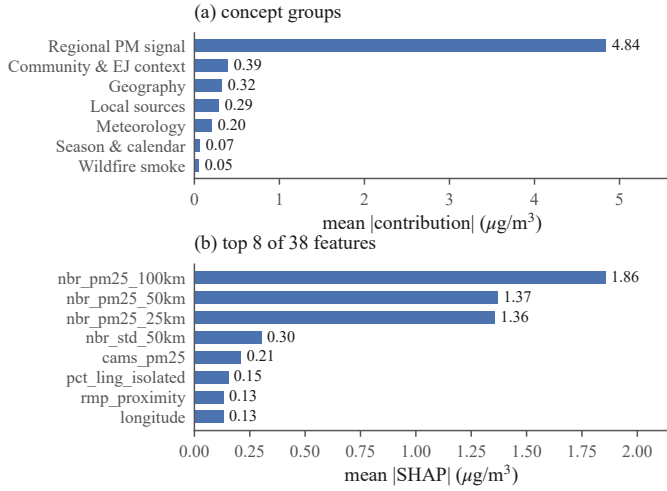


Fig. 1. Global importance over the 6,000-row sample. (a) Mean absolute concept-group contribution; the regional PM signal exceeds every other group by an order of magnitude. (b) The top individual features are the three neighbor means, neighbor dispersion, and CAMS surface PM2.5.

(April 3, 2024) among days with at least 150 reporting sensors. Local cases follow fixed rules: the highest-reading smoke-day row predicted within 35% of its observation, and the worst underprediction above  $40 \mu\text{g}/\text{m}^3$ . All pipeline outputs reproduce byte-identically, and every analysis number in this paper is regenerated by verification scripts released with the code.

#### IV. RESULTS

##### A. The model is a spatial nowcaster

Fig. 1 shows global importance. The regional PM signal group averages  $4.84 \mu\text{g}/\text{m}^3$  of absolute contribution per prediction; the next group, community and EJ context, averages  $0.39 \mu\text{g}/\text{m}^3$ , and the dedicated smoke tier is last at  $0.05 \mu\text{g}/\text{m}^3$ . The three neighbor-mean features are individually the three most important inputs ( $1.86$ ,  $1.37$ , and  $1.36 \mu\text{g}/\text{m}^3$  at 100, 50, and 25 km). Whatever else the ensemble encodes, its predictions are first and foremost a spatial average of what nearby sensors already report. The ordering is robust: a sensor-clustered bootstrap reproduces the full group ranking in 89.3% of 1,000 resamples, and grouped permutation importance orders the seven groups identically (rank correlation 1.0).

##### B. Feature response and semantics

Dependence analysis (Fig. 2) sharpens this picture in five ways.

1) *Near-linear regional response*: The 50 km neighbor mean maps onto its attribution almost monotonically (Spearman  $\rho = 0.994$ ), with slope  $0.26 \mu\text{g}/\text{m}^3$  per  $1 \mu\text{g}/\text{m}^3$  of neighbor mean below  $30 \mu\text{g}/\text{m}^3$ , then flattens: recomputed exactly over the corpus, the median contribution is  $+6.3 \mu\text{g}/\text{m}^3$  for neighbor means between  $30$  and  $40 \mu\text{g}/\text{m}^3$  and  $+8.0 \mu\text{g}/\text{m}^3$  (CI  $[7.93, 7.97]$ , all 860 qualifying rows) above  $40 \mu\text{g}/\text{m}^3$ . The flattening is consistent with the exclusion of training days above  $75 \mu\text{g}/\text{m}^3$ : the model has little headroom to extrapolate events it never saw.

2) *The smoke tier is nearly redundant, and not monotone*: Mean attribution of the HMS tier, in  $\mu\text{g}/\text{m}^3$ , is  $-0.044$  (none) and  $+0.060$  (light) on the sample,  $+0.110$  over 2,500 recomputed medium-tier rows (CI  $[0.108, 0.111]$ ), and  $+0.097$  over all 431 heavy-tier rows (CI  $[0.094, 0.101]$ ). Heavy plumes contribute less than medium ones (bootstrap difference CI  $[0.008, 0.016] \mu\text{g}/\text{m}^3$ ), though the  $0.012 \mu\text{g}/\text{m}^3$  gap is negligible against an RMSE of 4.22. Only 431 of 285,798 training rows (0.15%) carry the heavy tier, and on those rows mean observed PM2.5 is lower than on medium-tier rows ( $11.6$  versus  $16.6 \mu\text{g}/\text{m}^3$ ), a reminder that an analyst-drawn plume overhead does not guarantee surface-level smoke. The model nearly ignores a feature whose evidential value its training data did not support.

3) *Proximity features invert at the extreme*: Traffic proximity attribution is flat over most of its range, then turns negative past roughly the 85th percentile: mean  $-0.36 \mu\text{g}/\text{m}^3$  beyond that point,  $-1.15 \mu\text{g}/\text{m}^3$  over the top two percent, and single predictions as low as  $-3.6 \mu\text{g}/\text{m}^3$ . Diesel PM proximity behaves the same way at smaller magnitude. Extreme proximity marks dense urban cores whose sensor-rich surroundings already carry the level; the model uses it as a downward geographic correction, not an emissions signal. A naive reading, that traffic lowers pollution, would invert the causal reality.

4) *EJ context is small, monotone in binned median, and strictly correlational*: Attribution rises with percent linguistically isolated ( $\rho = 0.622$ , sensor-clustered bootstrap CI  $[0.52, 0.72]$ ) and percent people of color ( $\rho = 0.628$ , CI  $[0.53, 0.72]$ ); above its 70th percentile, linguistic isolation is associated with binned-median attributions of up to  $+0.49 \mu\text{g}/\text{m}^3$ . These are learned associations between demographics and measured PM2.5 under the model's other inputs. They must never be presented as demographic causation, and we do not compute counterfactuals over them.

5) *Interactions stay inside the regional group*: SHAP interaction values (a 300-row subsample, seed 42; random forest and CatBoost combined with renormalized weights covering 99.2% of the blend, LightGBM at weight 0.008 omitted) rank every leading pair within the regional signal group: the neighbor means interact at  $0.50$ ,  $0.46$ , and  $0.28 \mu\text{g}/\text{m}^3$  mean absolute strength, followed by neighbor-dispersion pairs. The strongest pair that crosses group lines, diesel proximity with linguistic isolation, is an order of magnitude weaker at  $0.05 \mu\text{g}/\text{m}^3$ . Regional dominance is not an artifact of additive attribution.

##### C. Two days, one mechanism

On May 27, 2024 (Fig. 3), with 98.7% of sensors under an analyst-drawn plume and a statewide mean reading of  $37.0 \mu\text{g}/\text{m}^3$ , the model predicts a mean of  $36.1 \mu\text{g}/\text{m}^3$ , and the regional signal group contributes  $+24.8 \mu\text{g}/\text{m}^3$  of that on average. The smoke tier's largest contribution anywhere in the state is  $+0.31 \mu\text{g}/\text{m}^3$ . On April 3, 2024 (statewide mean  $1.03 \mu\text{g}/\text{m}^3$ ), the regional group pulls predictions down by  $7.6 \mu\text{g}/\text{m}^3$  on average from the  $9.72 \mu\text{g}/\text{m}^3$  baseline. Smoke

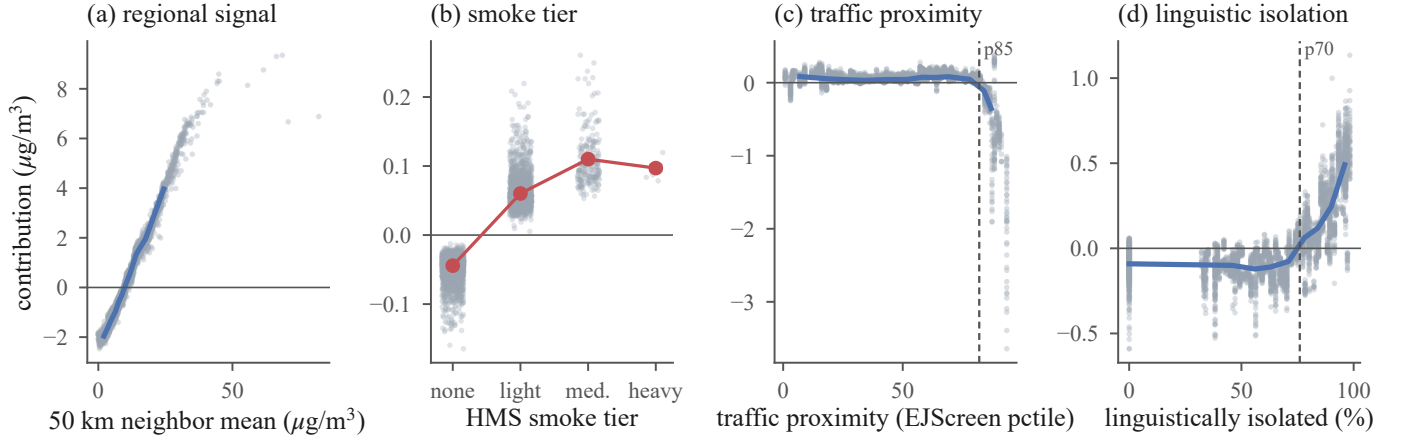


Fig. 2. Feature dependence on the 6,000-row sample (gray points; blue line is the 12-quantile binned median). (a) Contribution of the 50 km neighbor mean is near-linear then flattens above  $40 \mu\text{g}/\text{m}^3$ . (b) HMS smoke tier: red markers are exact means over all 431 heavy-tier rows and 2,500 medium-tier rows with bootstrap 95% intervals; heavy contributes less than medium. (c) Traffic proximity is inert until roughly the 85th percentile, then turns negative. (d) Linguistic isolation rises monotonically past its 70th percentile.

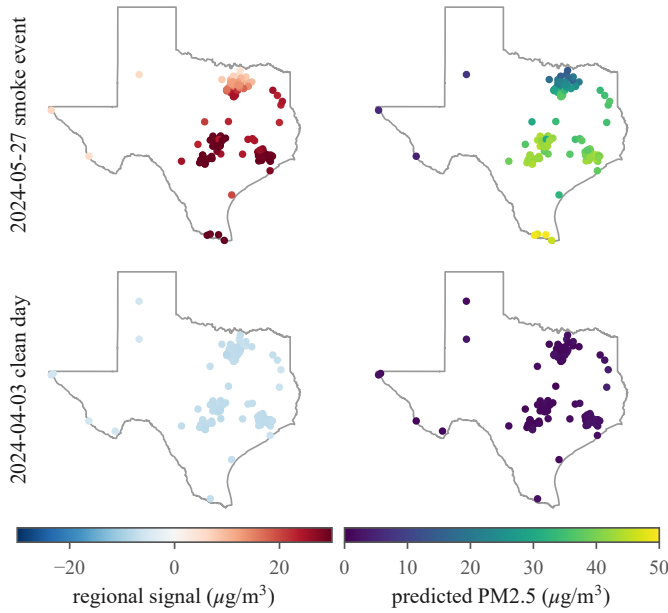


Fig. 3. Every reporting sensor explained on the strongest smoke day (top, 156 sensors, 98.7% under a plume) and the cleanest day (bottom, 154 sensors): regional-signal contribution (left) and predicted PM2.5 (right). The event is carried almost entirely by the regional signal ( $+24.8 \mu\text{g}/\text{m}^3$  statewide mean) while the smoke tier never exceeds  $+0.31 \mu\text{g}/\text{m}^3$  anywhere in the state.

information reaches the model through the neighbor network, not the smoke feature: both days are settings of one nowcasting mechanism.

#### D. Explaining a success and a failure

Fig. 4 decomposes two individual predictions. On the smoke day, sensor 217461 read  $71.1 \mu\text{g}/\text{m}^3$  and the model predicted  $49.6 \mu\text{g}/\text{m}^3$ , with  $+34.6 \mu\text{g}/\text{m}^3$  from the regional signal and  $+0.29 \mu\text{g}/\text{m}^3$  from the smoke tier; even this detection underpredicts, consistent with the capped training range and the saturating neighbor response. On November 19, 2024, sensor 242357 read  $75.0 \mu\text{g}/\text{m}^3$ , at the retention boundary

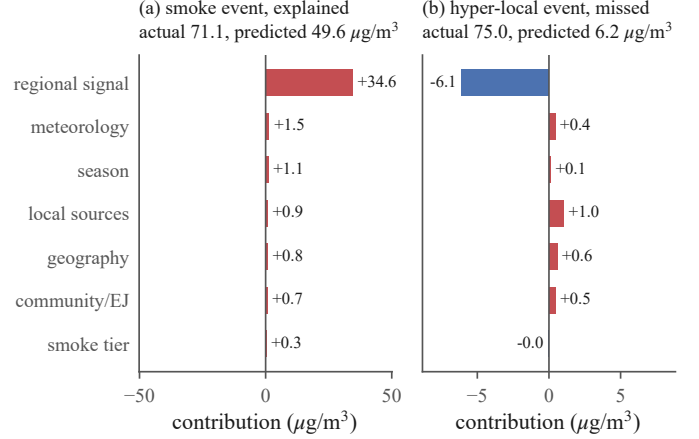


Fig. 4. Exact concept-group decomposition of two predictions. (a) A detected smoke event is carried by the regional signal; the smoke tier adds  $+0.3 \mu\text{g}/\text{m}^3$ . (b) The worst miss in the corpus: neighbors saw nothing, and the regional signal suppresses the prediction below the baseline.

itself, while every neighbor mean sat below  $1.8 \mu\text{g}/\text{m}^3$ ; the model predicted  $6.2 \mu\text{g}/\text{m}^3$ . The explanation shows why: the regional group contributed  $-6.1 \mu\text{g}/\text{m}^3$ , the three neighbor means  $-2.5$  to  $-1.7 \mu\text{g}/\text{m}^3$  each. Nor is the case unique: 46 corpus rows read at least  $40 \mu\text{g}/\text{m}^3$  with every neighbor mean below  $5 \mu\text{g}/\text{m}^3$ ; the regional group is negative on all 46 (mean  $-5.2 \mu\text{g}/\text{m}^3$ ), and the highest prediction among them is  $9.1 \mu\text{g}/\text{m}^3$  against a mean observation of  $54.1 \mu\text{g}/\text{m}^3$ . Rows above  $75 \mu\text{g}/\text{m}^3$  were excluded (241 pre-cap), so these figures lower-bound the blind spot. The features that make the model accurate on regional events actively vote against hyper-local ones; no reweighting of the same features can fix what the network cannot see.

## V. DISCUSSION

### A. For model builders

The smoke tier could be replaced, not retuned: candidate representations include plume density, smoke age, or lagged

fire radiative power; only 431 heavy-tier rows support the current one. The saturating neighbor response ties directly to the  $75 \mu\text{g}/\text{m}^3$  training cap, so raising the cap (or modeling exceedances separately) is a measurable next experiment.

### B. For network operators

The miss case turns an error metric into a map, with one caution: sign alone cannot separate clean surroundings from unsensed ones (the clean day’s statewide mean,  $-7.6 \mu\text{g}/\text{m}^3$ , is as negative as El Paso’s annual  $-7.0 \mu\text{g}/\text{m}^3$ ). Sparsity separates them: the El Paso sensor averages 0.8 same-day neighbors within 50 km against a statewide 27, while densely sensed East Texas runs  $+9.3 \mu\text{g}/\text{m}^3$ . Persistently negative regional means in sparse neighborhoods mark where predictions rest on remote evidence (distance to the nearest sensor reaches 164 km), and those are the places where the platform’s placement optimizer should spend new sensors; the explanation layer supplies that prioritization for free.

### C. For communicating attributions

SHAP explains the model, not the atmosphere. The traffic inversion shows a raw attribution contradicting physical causality; the EJ attributions show a small, systematic association that could be misread causally. Any deployment of these explanations to a public dashboard needs the concept-group framing, the correlational disclaimer, and suppression of counterfactuals on demographic features as defaults, not footnotes.

### D. Limitations

The target is the uncorrected ATM channel, so absolute levels are not comparable to regulatory monitors without a correction such as that of Barkjohn et al. [1]; the explanations inherit whatever bias that channel carries. Days above  $75 \mu\text{g}/\text{m}^3$  are absent from training, so extreme-event behavior describes a model that never saw the worst days, and the failure analysis lower-bounds the blind spot. LOSO  $R^2$  characterizes held-out sensor locations; accuracy at unsensed tracts is unquantified, and the miss case suggests it degrades exactly there. The EJ attributions are further conditioned on a network whose placement is itself demographically uneven [2], so their sign and size partly reflect where sensors exist. Tree-path-dependent attribution spreads credit among correlated features, so within-group rankings are less certain than group totals. Explanation statistics describe the training frame, not the serving distribution. The findings describe one model over one state and five years; the method applies unchanged to any convex blend of tree models.

## VI. CONCLUSION

We converted a deployed, accuracy-validated PM2.5 mapping system into one whose every prediction can be read: an exact blend of per-model TreeExplainer attributions, folded into seven concept groups, reconstructs each output to within  $10^{-6} \mu\text{g}/\text{m}^3$ . The reading is unflattering in places. The model is a spatial nowcaster that nearly ignores its smoke flag, inverts

proximity features at the urban extreme, and structurally cannot see events its sensor network does not cover. None of this is visible in  $R^2$ , and all of it matters for how the tool should be used, improved, and communicated. Explaining 6,000 predictions takes 20 minutes on a desktop CPU and one prediction a fraction of a second, so attaching exact, plainly narrated explanations to the live platform is the immediate next step.

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